How sure is your house-price model?



A hands-on sesssion on uncertainty quantification

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Today: a UQ workflow for regression

A housing case, two exercises in R, and a discussion of different approaches to UQ.

Aleatoric and epistemic uncertainty in a simulation

Outcome variation and uncertainty about what we have learned.

1. Prediction intervals

What could a new house sell for?
Build, check and calibrate intervals.

2. Model sensitivity

How much does the fitted median depend on the training data?
Compare whole-forest refits.

Take home: a prediction-interval workflow and a separate model diagnostic.

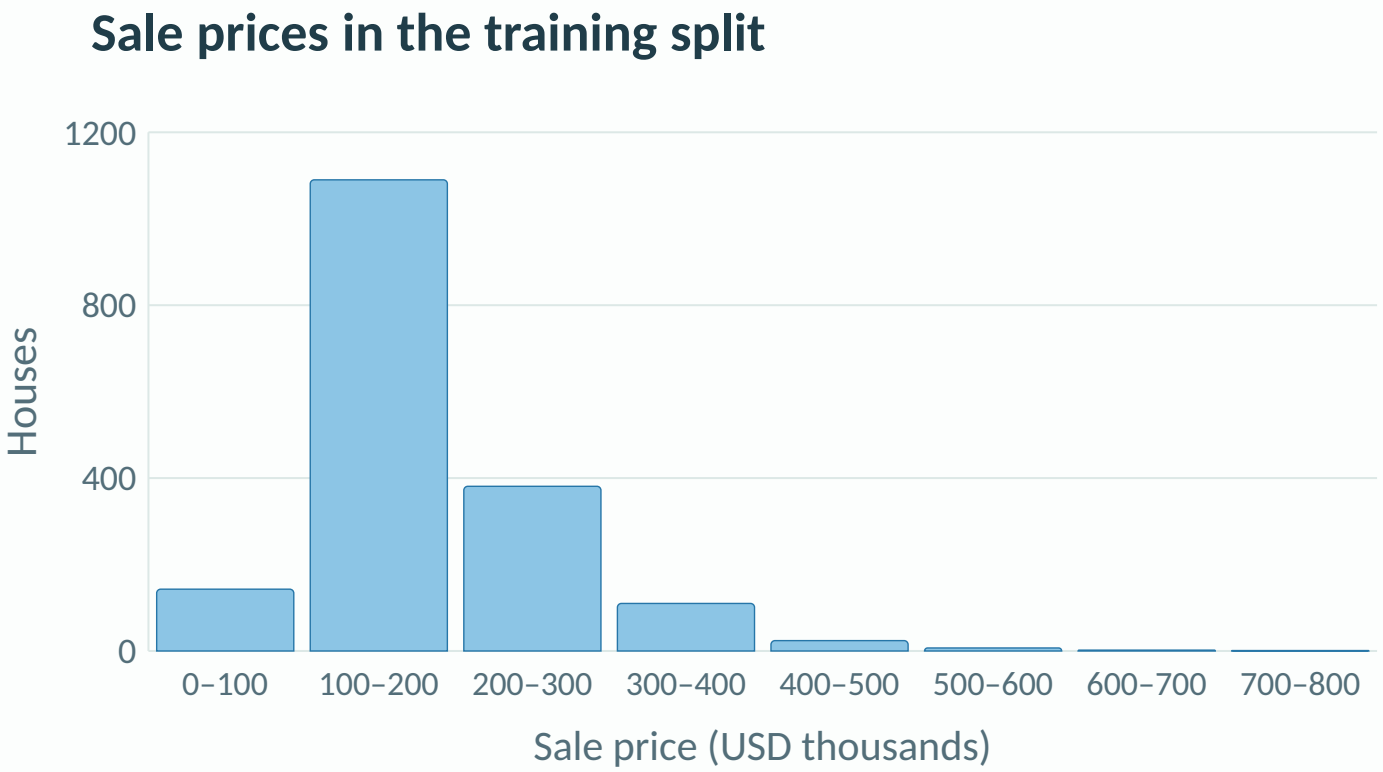
Ames, Iowa

A real regression task with familiar property information

2,930

Residential sales
Ames, Iowa, 2006–2010

Predict price from size, age and neighbourhood.



Your first price (gu)es(s)timate

House A:

- 114 m^2 living area
- 3 bedrooms
- Built in 1956
- Edwards neighborhood

Your price?
Your 90% range?

For scale: the median training sale price is $159,217.

Write your estimate and range. We will return to this house.

House A: the forest's point prediction

$132,000

Quantile forest: predicted median sale price

What would you want to know before relying on it?

Aleatoric and epistemic uncertainty

Aleatoric

Prices vary even when we know
the outcome distribution.



One known distribution

Epistemic

We are uncertain about
the outcome distribution itself.



Several possible distributions

The distinction depends on the information and modelling assumptions we use.

First: a simulation with known uncertainty

A simple linear model makes the distinction visible.

50 vs 500 simulated training sales

Same house features, population and outcome noise.

- Does the true outcome range change?
- Do the fitted medians become more stable?

More training data improves estimation

Same simulated house, with 20 training data sets and model fits at each sample size.

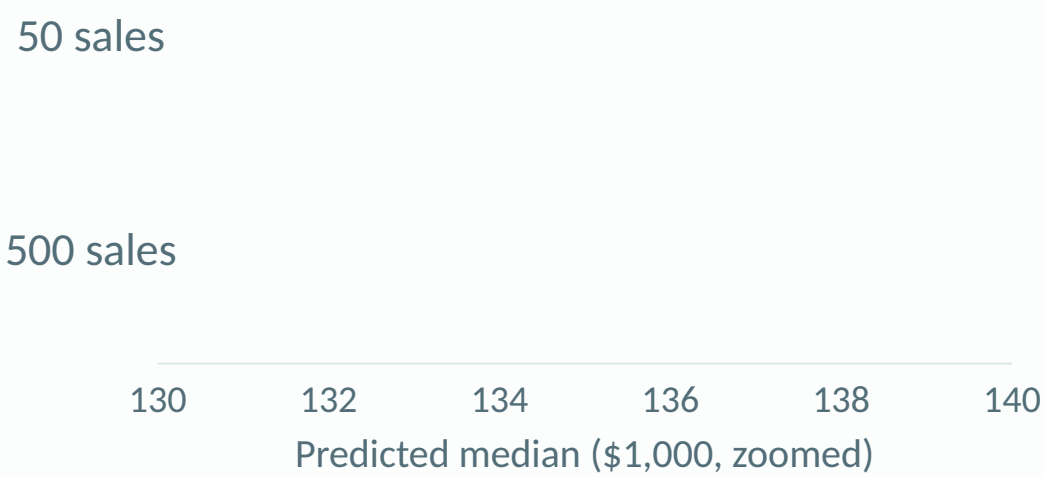
Aleatoric uncertainty

True 90% sale-price range



Epistemic: learning the median

Each dot: one model's predicted median



Predictions become more stable, but outcome variability stays the same.

1. Prediction intervals for sale prices

What could this house sell for?

- Estimate the 5th and 95th price percentiles
- Report a range for a future sale

How we check the range

- Measure coverage and width
- Calibrate using reserved data
- Check performance on new cases

Our output: a prediction interval for an individual sale price.

Training, calibration and final evaluation

2,930 houses in one fixed split



Training

Fit preprocessing and the forest

Calibration

Adjust interval endpoints

Test

Check coverage and width

How the forest builds a prediction interval

QRF estimates price percentiles using sales grouped by the forest.

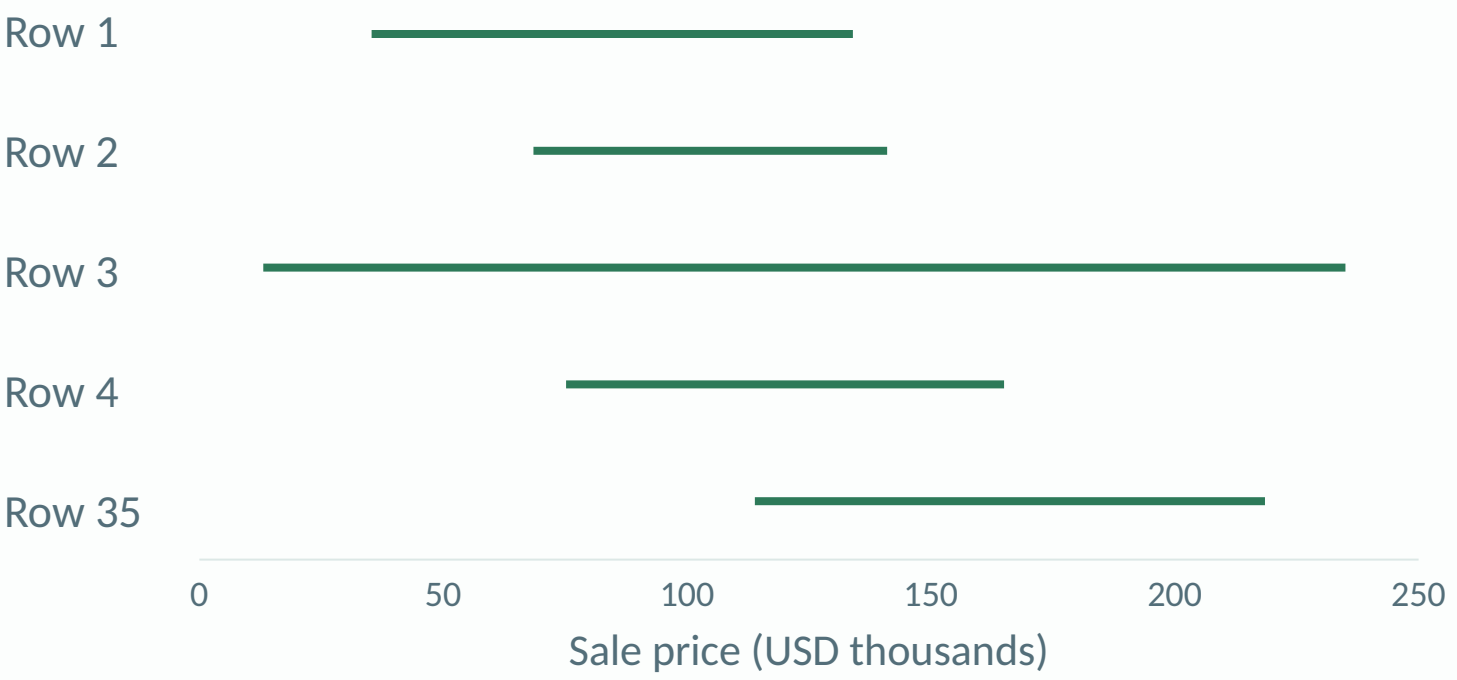


A 90% prediction interval for this house's sale price

90% is the target coverage: we will check it on held-out houses.

Checking QRF intervals against sale prices

Five selected test houses (the exercise uses all 586).



Green: QRF interval Orange dot: covered sale Orange cross: missed sale

Coverage

4 / 5 = 80%

Observed prices inside their intervals

Width of row 1

$98,689

$134,000 – $35,311

How could we raise coverage? What would happen to width?

Exercise 1: the data you will evaluate

head(data.frame(price = y, intervals), 3)

Row	Observed price	Lower	Median	Upper
1	**$72,500**	$35,311	$99,000	$134,000
2	**$113,000**	$68,500	$118,000	$141,025
3	**$81,000**	$13,100	$105,450	$235,000

QRF endpoints: lower = 5th percentile, upper = 95th.

Is the first observed price inside its interval?

Exercise 1: do the QRF intervals work?

You're provided with 90% prediction intervals for sale prices on 586 test houses.

https://github.com/Camiling/NORA-UQ-workshop

Get the files from GitHub. Open nora_uq_workshop.Rproj.

Start here (5 min)

- Run 00_setup.R
- Open 01_intervals.R
- Use the included predictions

Work in pairs (17 min)

- Calculate coverage and mean width
- Inspect the interval plot
- Discuss usefulness for a buyer

Need a guide? Open START_HERE.html

Solution: solutions/01_intervals_solution.R

The QRF intervals covered 96.2% of test sales

Target coverage: 90%.

96.2%

Observed coverage

$112,378

Mean interval width

From predictions to coverage and width

```
covered = y >= intervals$lower & y <= intervals$upper
coverage = mean(covered)
mean_width = mean(intervals$upper - intervals$lower)
```

How useful are these intervals for a housing decision?

QRF prediction intervals for three houses

House A sold for $135,000. Did your opening range include it?



Green: QRF interval Blue cross: predicted median Orange dot: sale price

Calibration of the prediction intervals

Conformalized quantile regression (CQR), using reserved calibration data

The same target: a range for a future sale price.
Use 586 calibration sales to adjust the endpoints.

- Compare calibration prices with their QRF intervals
- Calculate the adjustment for the 90% coverage target
- Evaluate coverage and width on the test houses

Run together: 03_calibration_demo.R

Will calibration widen or narrow the intervals?

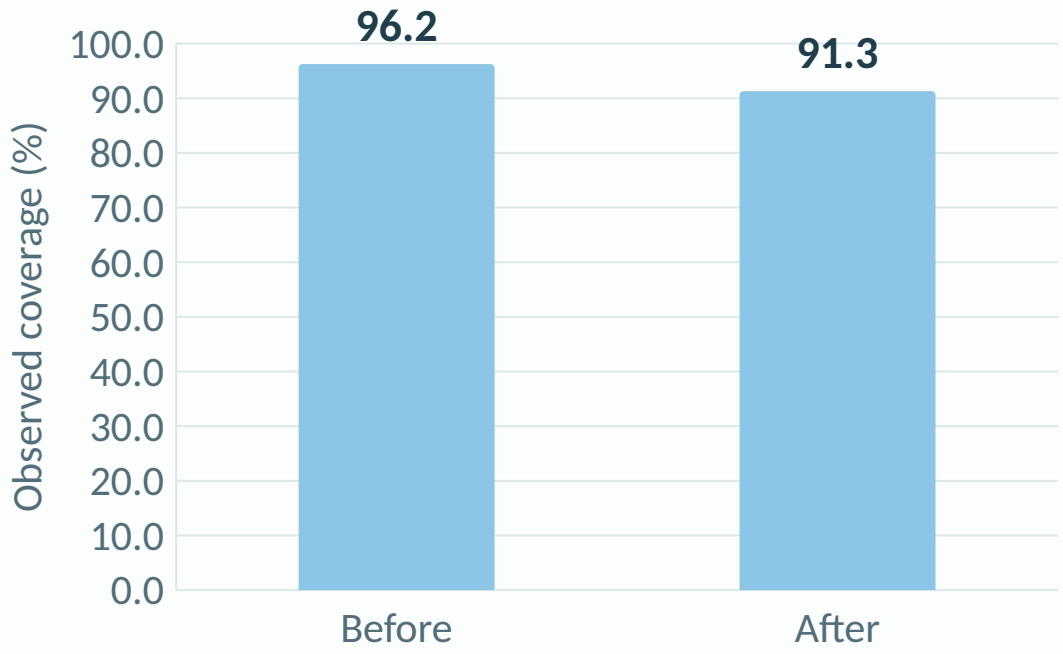
[lower, upper]

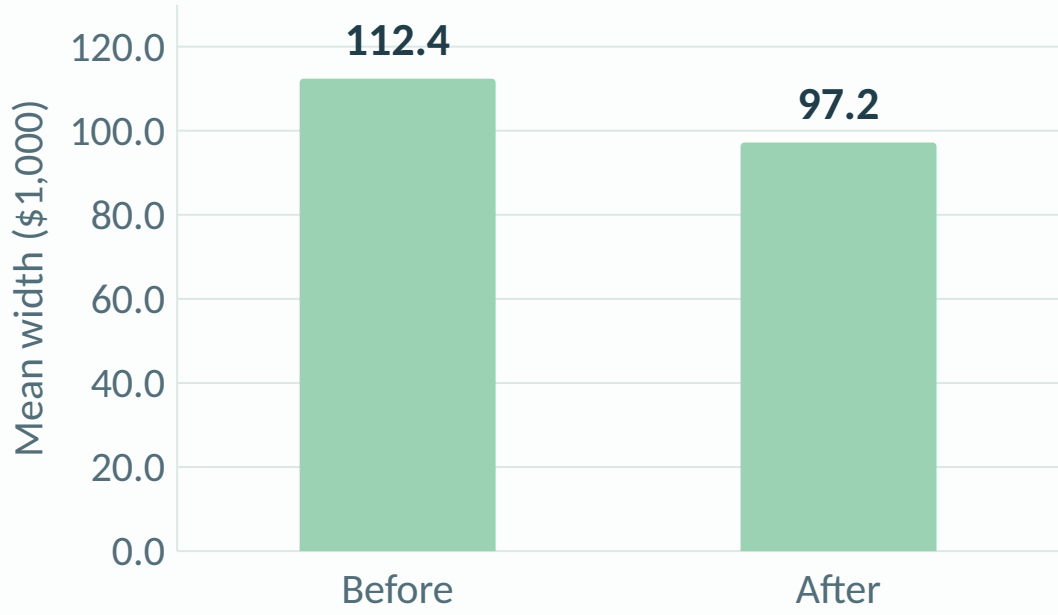


[lower-adjustment, upper+adjustment]

Calibrated 90% intervals were narrower in this case

Before: QRF intervals. After: the conformal adjustment.

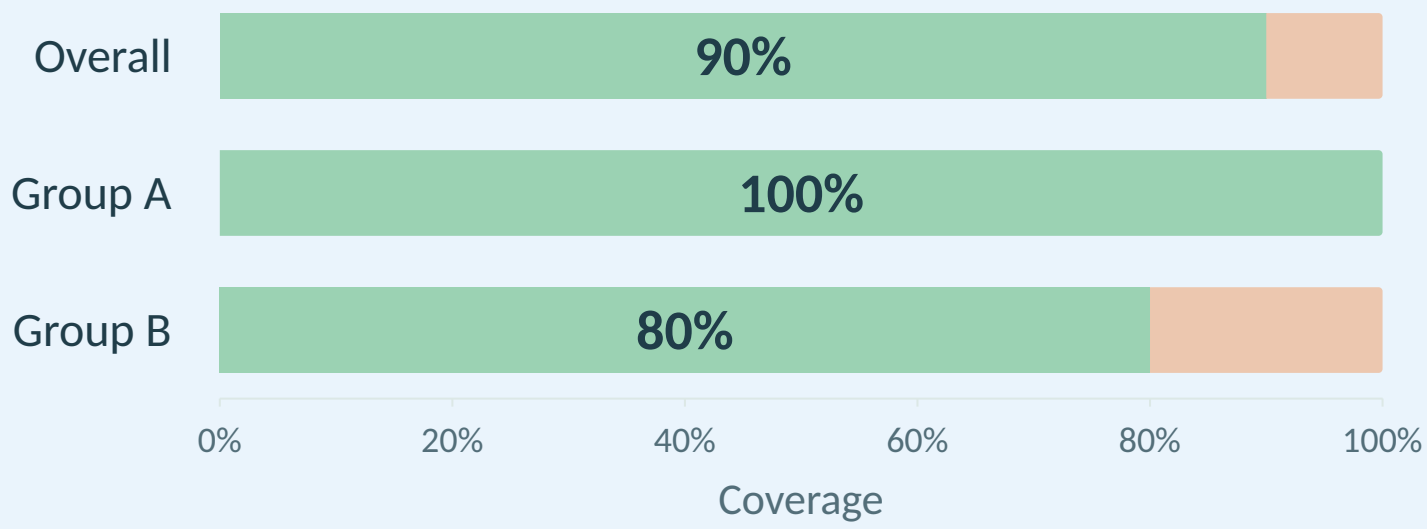




Note: overall coverage can hide a subgroup gap

Conformal prediction targets marginal coverage under exchangeability.

Illustration: 100 cases split into two groups of 50



90 of 100 covered overall

40 of 50 covered in Group B

Which groups would you check in your application?

Coffee, questions, and a fresh look

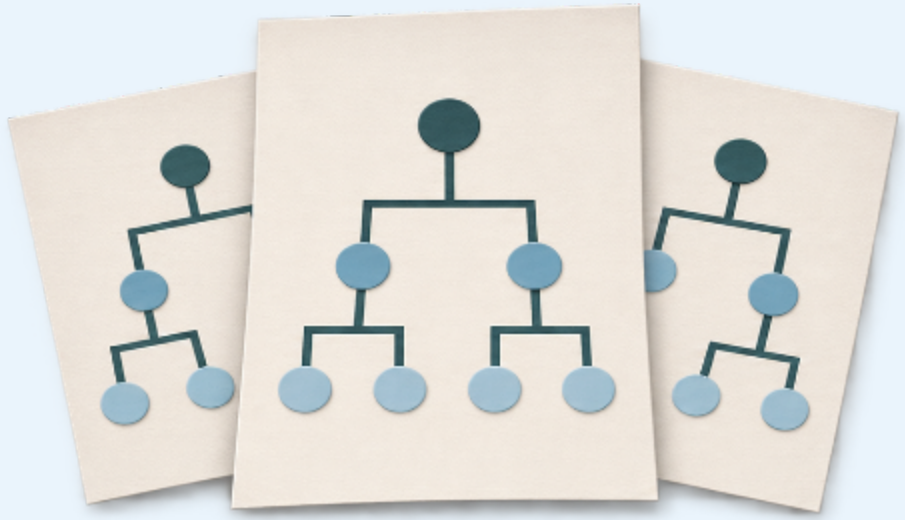
Back at 14:15

Next: how sensitive is the fitted median to the training data?

2. Sensitivity of the learned median

How much does our fitted median depend on the specific training sample?

Keep forest size fixed



Vary sampled training sales



Diagnostic: resample sales, refit 100-tree forests, compare predicted medians.

Resampling sales changes the fitted median

Illustration with House B: resample training sales, refit a forest, predict its median.
Repeat 20 times at each sample size.



Circles: 300 rows. Diamonds: 1,500 rows.

Ensemble SD (refit SD): spread of 20 fitted medians for one house.

Exercise 2: sensitivity of the fitted median

20 training data resamplings and forest refits at each sample size.
Predict on the same 100 target houses.

https://github.com/Camiling/NORA-UQ-workshop

Open 02_refits.R.

What you compare

- 300 versus 1,500 sampled sales
- SD across 20 predicted medians
- Smaller SD means greater stability

Work in pairs (17 min)

- Summarize SDs at each sample size
- Count houses with lower SD for the larger sample size
- Explain what this SD measures

Solution: solutions/02_refits_solution.R

Bring back two median SDs and a fraction.

Most houses had more stable median predictions

Refit SD: spread of 20 predicted medians for one house.



87 / 100

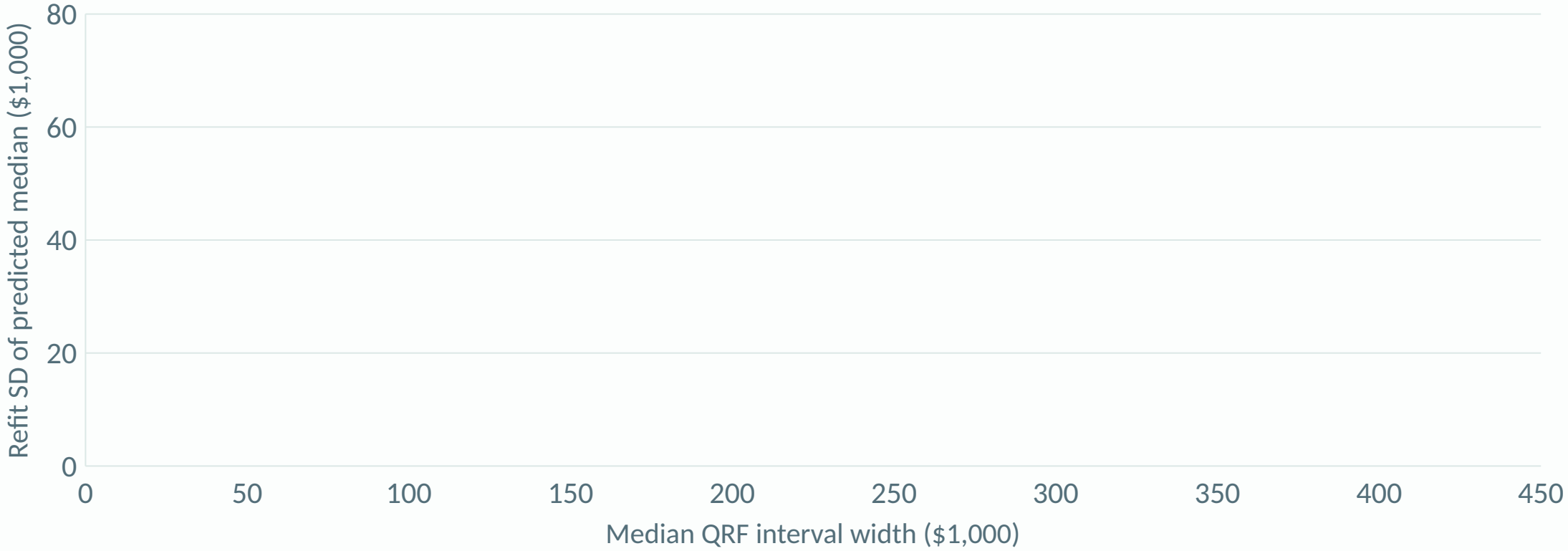
lower SD with 1,500 rows than with 300 rows

median(small$prediction_sd)
median(large$prediction_sd)
mean(large$prediction_sd < small$prediction_sd)

What could all these fits still miss?

Prediction-interval width vs ensemble SD - do they agree?

Each dot: one house, summarized over 20 refits with 1,500 rows.



A wide interval can coexist with a stable fitted median.

Limitation: stable predictions can share blind spots

Every refit uses the same recorded features and the same 2006–2010 housing market.

- What if an important property detail is missing?
- What if we predict in a different market?

Agreement among fits still leaves these questions open.

Summarizing: how we compute the two quantities

Prediction interval

- Fit one quantile forest
- Read its 5th and 95th price percentiles
- Calibrate with reserved sales

Ensemble SD

- Refit 20 forests on resampled data
- Predict the same house each time
- Take the SD of the 20 medians

Which quantity should I use?

Prediction interval

Need a range for a future sale?



A range of possible sale prices.

Ensemble SD

Need to check sensitivity to the training data?



Spread of the fitted medians.

Different summaries answer different questions

- Prediction interval: a range for a future sale price
- Ensemble SD: sensitivity of the fitted median

An explicit aleatoric–epistemic split
requires modelling assumptions.

You can explore this in 07_model_assumptions.R (optional).

References and workshop files

- **Hüllermeier & Waegeman (2021)**
  Aleatoric and epistemic uncertainty in machine learning
- **Meinshausen (2006)**
  Quantile Regression Forests
- **Romano, Patterson & Candès (2019)**
  Conformalized Quantile Regression
- **De Cock (2011)**
  Ames housing data — Journal of Statistics Education, 19(3)

https://github.com/Camiling/NORA-UQ-workshop

On GitHub: exercises, worked solutions and optional follow-up scripts.